

Evaluation of BeeGFS on a RISC-V Cluster for HPC Storage Infrastructure

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Agenda

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- Related Work
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Introduction

- RISC-V ISA
 - RISC-V is an **open, modular** ISA enabling **flexible, customizable** processors for emerging HPC research.
- Parallel File Systems
 - Parallel file systems are designed to provide **high-throughput** and **concurrent access** to large datasets by distributing **data** and **metadata** across multiple storage nodes in **HPC** environments.
- BeeGFS Overview
 - BeeGFS is an **open source** parallel file system designed for scalable, high-performance storage and **efficient data management** in HPC environments.

Motivation

- RISC-V as an Open HPC Platform
 - RISC-V open and extensible ISA enabling **hardware–software** co-design for next generation HPC systems.
- HPC Storage is Performance Critical
 - Parallel file systems directly influence I/O performance and **application scalability**.
- HPC Storage on RISC-V is Unexplored
 - Native **BeeGFS** and **Lustre** deployments and behavior remain largely unexplored on RISC-V hardware.

Related Work

- Successfully ported, deployed and evaluated Lustre on a native RISC-V cluster.
- Demonstrated the feasibility of parallel file systems on emerging RISC-V platforms.
- Extended the study to BeeGFS to analyze alternative HPC storage solutions.

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Porting and Evaluation of Lustre on a RISC-V Cluster for HPC Storage Infrastructure

Authors: Surendra Billa, Rushikesh Jadhav, Yogeshwar Sonawane, Sanjay Wandhekar, Piyush Patle, Shaurya Rane [Authors Info & Claims](#)

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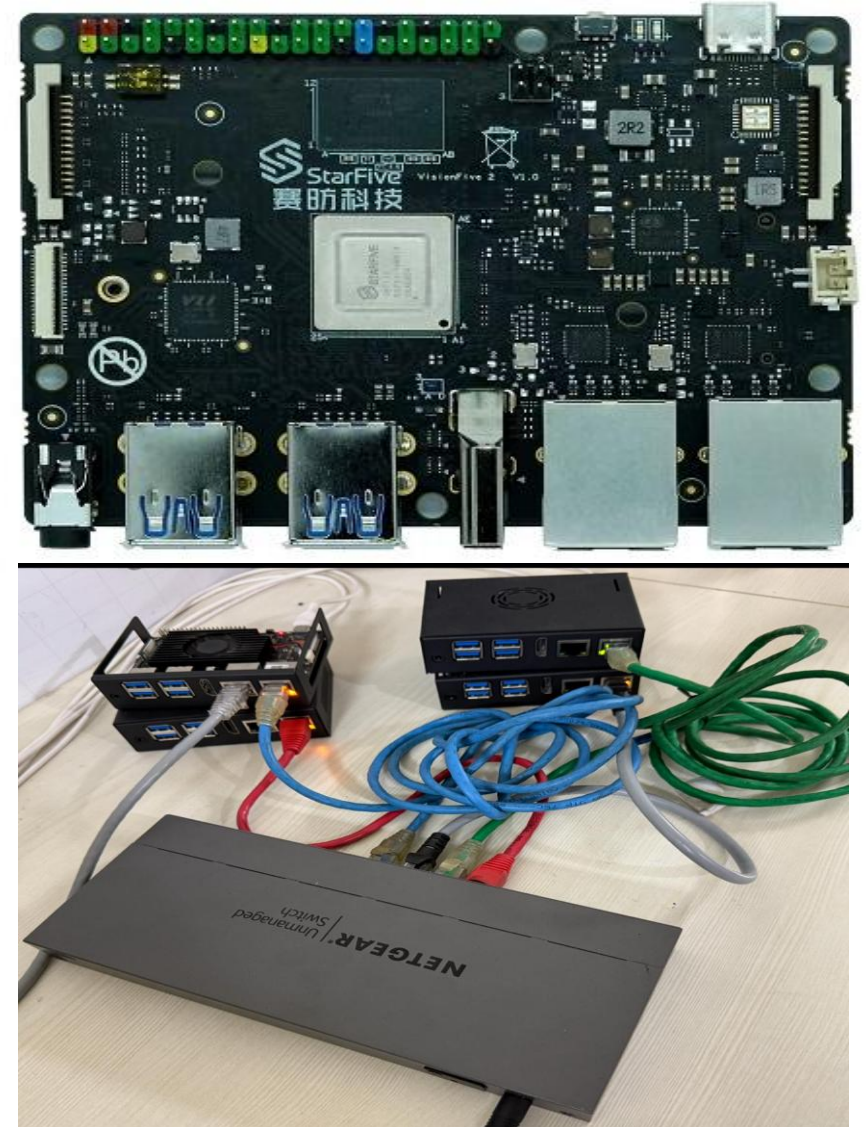
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Contributions

- Successful BeeGFS Compilation on RISC-V
 - **Compiled** and **deployed** the BeeGFS parallel file system on native RISC-V hardware.
- End-to-End Evaluation
 - Performed comprehensive **evaluation** covering deployment, **I/O throughput** and **metadata performance** using standard **HPC benchmarks**.
- ARM Baseline Comparison
 - Conducted a baseline study on an ARM based cluster with a comparable software stack and hardware to evaluate RISC-V performance characteristics.

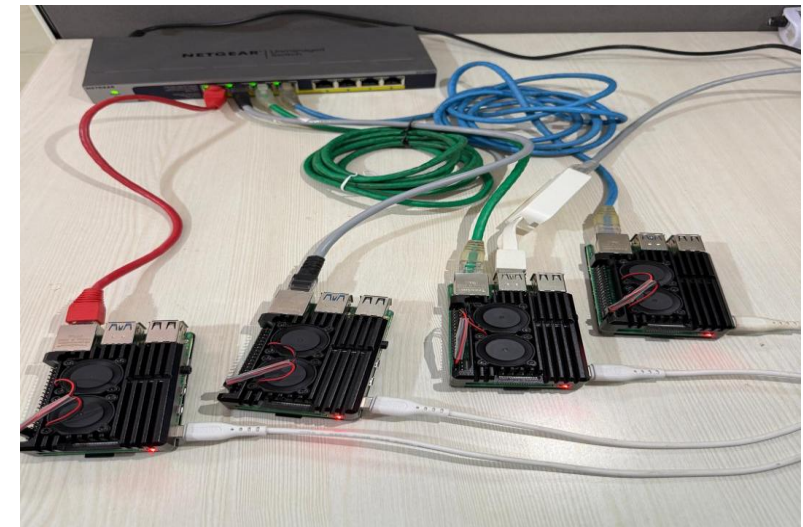
System Design – RISC-V

- 4-Node VisionFive2 Cluster
 - A compact RISC-V cluster built using four StarFive boards, each featuring a quad-core processor with 1.5 GHz.
- Fedora Linux 41 (RISC-V)
 - Using the standard RISC-V Fedora 41 Linux distribution as a stable software environment for HPC workloads.
- Gigabit Ethernet Interconnect
 - Nodes connected via 1 G ethernet star topology, ensuring reliable distributed storage.



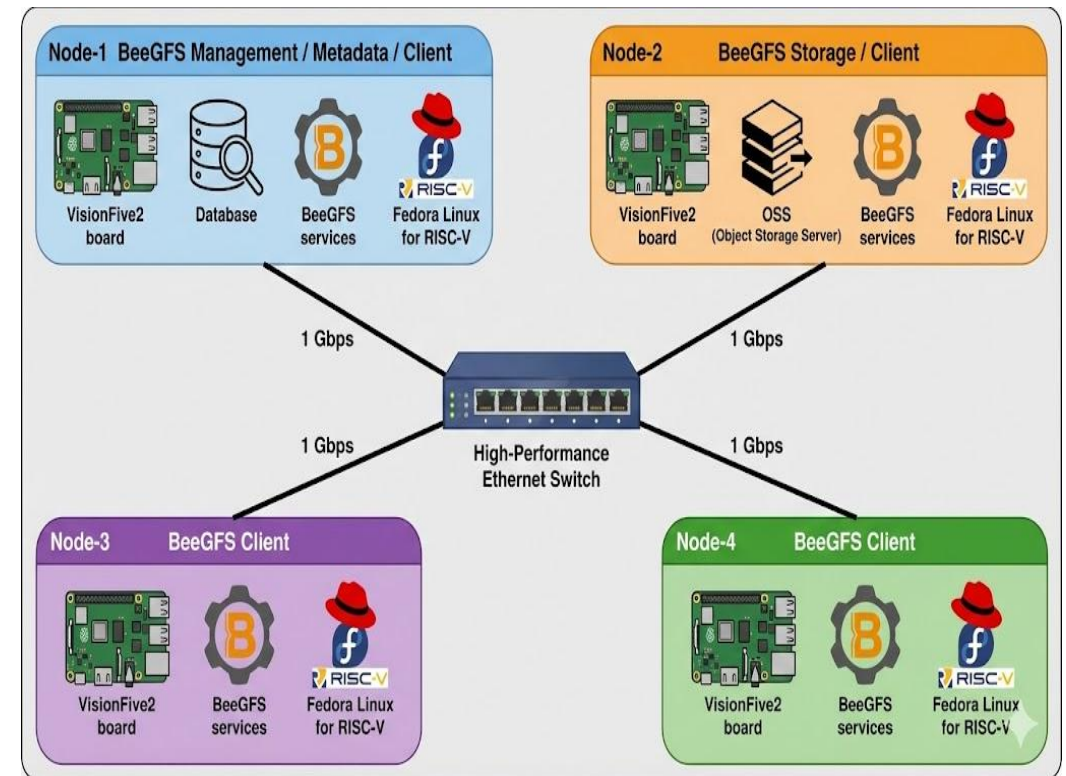
System Design – ARM

- 4-Node Raspberry Pi 4 B Cluster
 - A compact ARM cluster built using four raspberry Pi boards, each featuring a quad-core processor with 1.8 GHz.
- Fedora Linux 41 (ARM)
 - Fedora distribution tailored for ARM, providing a stable environment for HPC workloads.
- Gigabit Ethernet Interconnect
 - Nodes connected via 1 G Ethernet star topology, ensuring reliable distributed storage.



BeeGFS Architecture

- MGS / MDS (Management & Metadata Servers)
 - The MGS manages filesystem configuration; MDS handles metadata creation, permissions and namespaces.
- OSS with OSTs (Object Storage Servers)
 - OSS nodes store data across OSTs, enabling parallel read/write operations.
- Clients
 - Client nodes access filesystem concurrently, using MDS for metadata, OSS for data.



Compilation Process

- Downloaded **BeeGFS v8.2.2** source code and compiled it natively on the RISC-V platform.
- The build process completed successfully using the **native** RISC-V environment.
- No source code **modifications** were required during the compilation process on RISC-V.

Baseline Setup

- Raspberry Pi 4B Cluster
 - ARM based cluster used as a **reference platform** with comparable hardware.
- Fedora 41 Linux
 - Deployed the same version to ensure **OS level consistency** across platforms.
- Same Software Stack
 - Identical **BeeGFS version**, **kernel configuration** and **benchmarking tools** used for fair performance comparison.
- Cleared system caches between benchmark runs to ensure consistent and reliable performance measurements.

Benchmarks

- IOR
 - Evaluated sequential and parallel **read/write throughput** under varying block sizes, strip size, process counts and client configurations.
- MDTest
 - Measured metadata performance including file and directory creation, deletion and stat operations.

Benchmarks

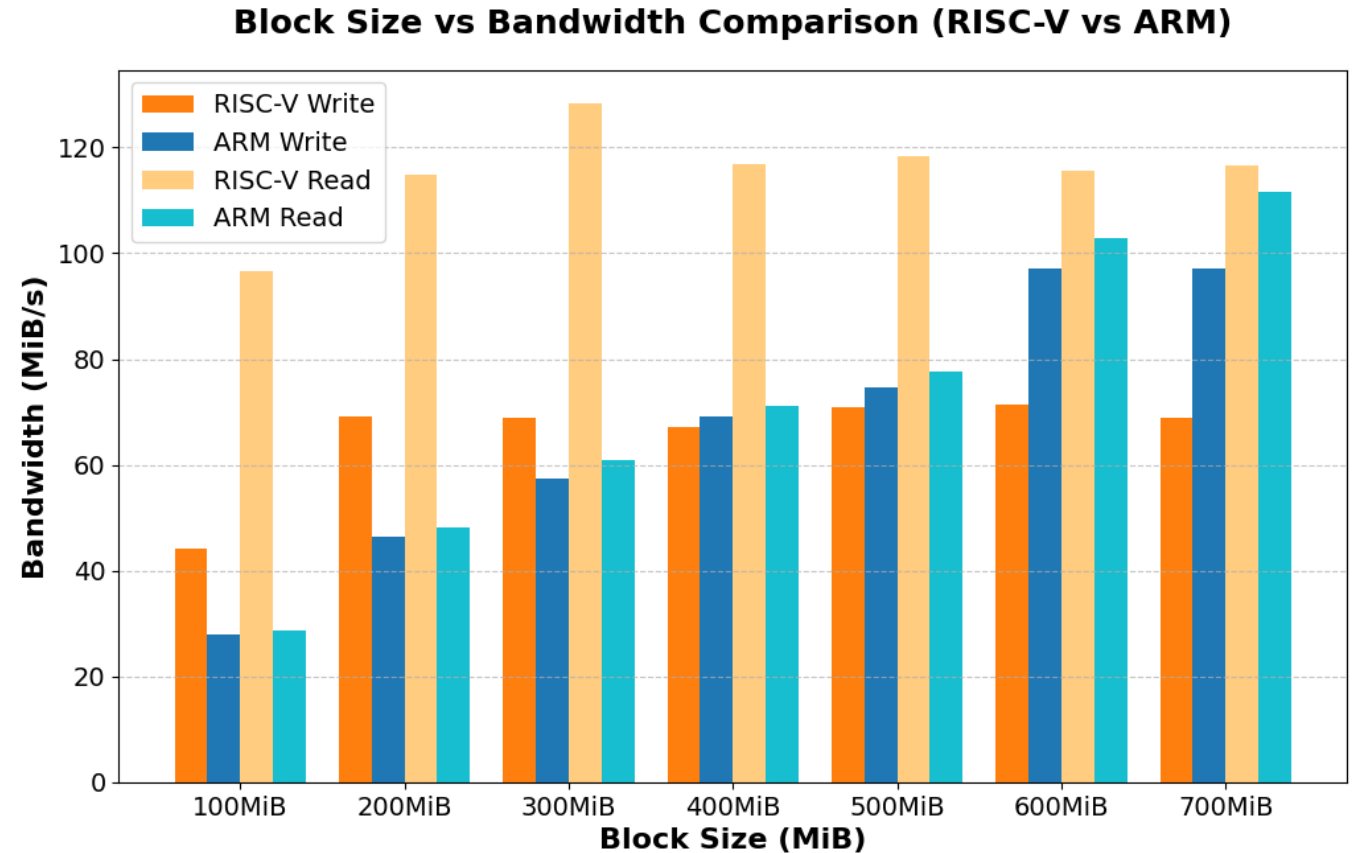
- IO500
 - Evaluated overall storage performance by combining IOR and MDTest workloads, measuring bandwidth, metadata operations and generating a unified IO500 score.
- Elbencho Benchmark
 - Evaluated high-throughput sequential I/O performance using **multi-threaded** workloads, measuring **bandwidth and IOPS** across the BeeGFS storage environment.

BeeGFS Performance Evaluation: IOR

- IOR enables systematic variation of I/O parameters, **process counts** and **client configurations** to study performance behavior and scalability.
- Our evaluation aims to understand beegfs performance under increasing I/O intensity and parallelism, which is critical for HPC workloads.
- Based on this approach, we organize the study into various IOR-based tests
 - Block Size
 - Stripe Size
 - Multi-Processing
 - Client count scaling

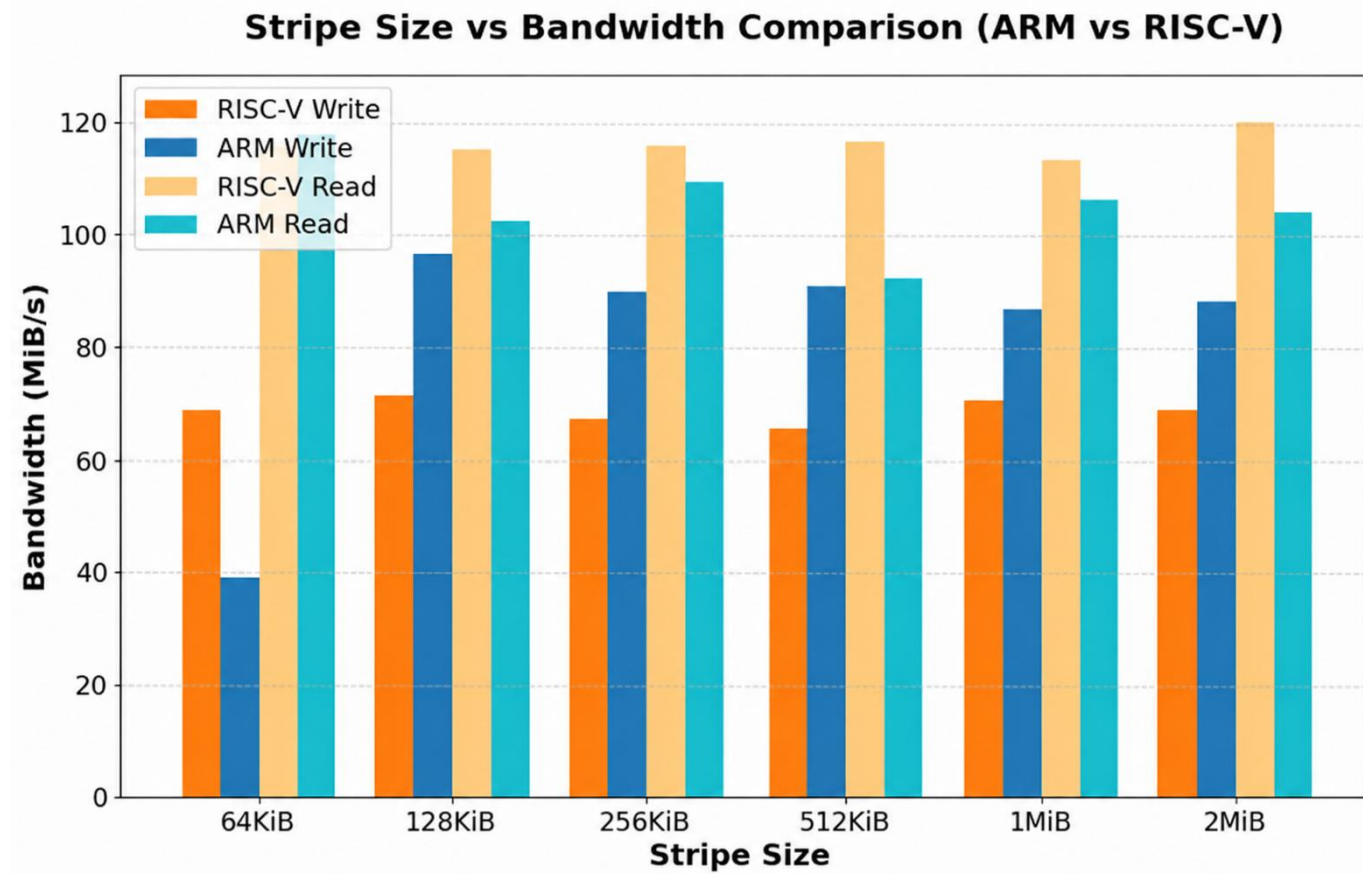
IOR – Block Size

- Scaling with Block Size
 - As the block size **increases**, I/O throughput **improves** and **stabilizes**
 - Demonstrating BeeGFS's efficient handling of large sequential accesses and effective utilization of the storage stack.



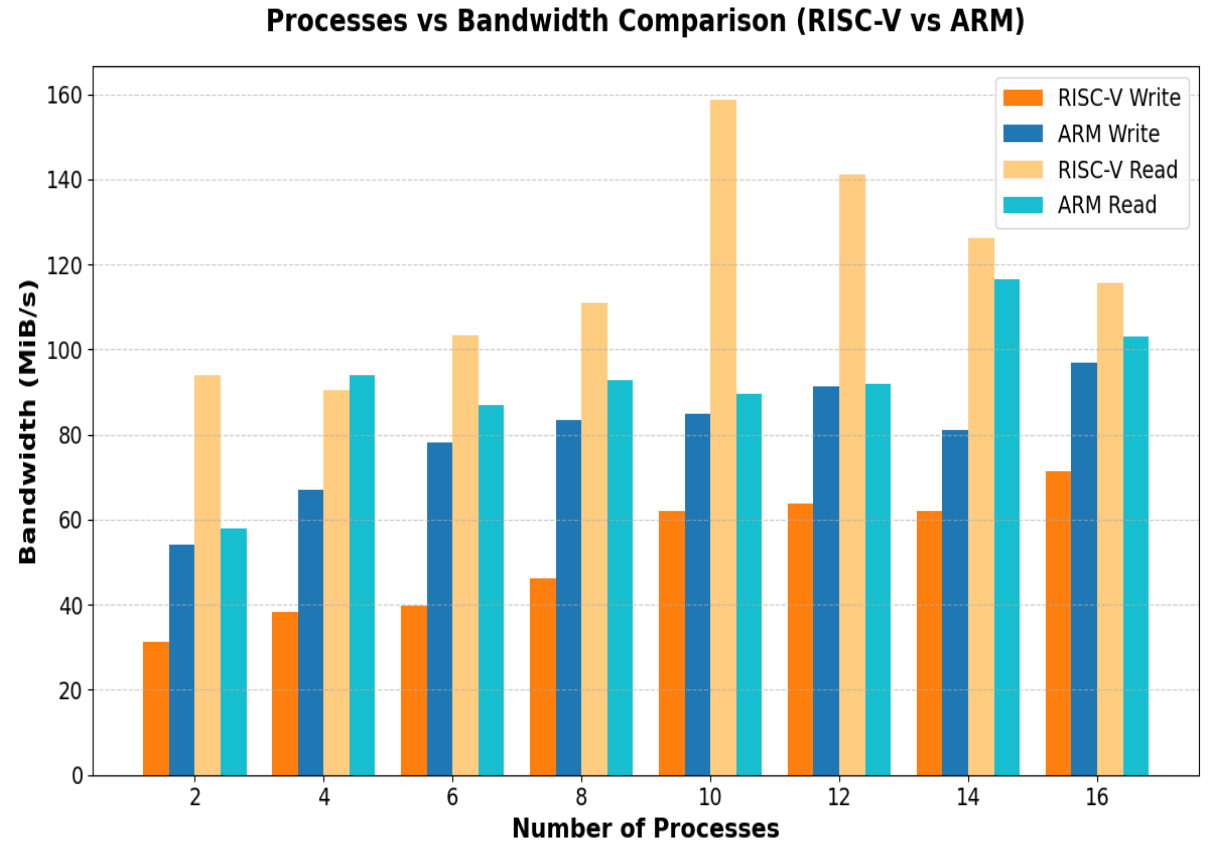
IOR – Stripe size

- **Configuration:** Stripe size varied from 64 KiB to 2 MiB with 16 MPI processes and a 600 MiB block size on BeeGFS.
- **Observation:** Write and read bandwidth remained relatively stable across different stripe sizes, with only minor performance variations.
- **Conclusion:** BeeGFS performance on RISC-V is largely unaffected by stripe size, demonstrating consistent I/O behavior across configurations.



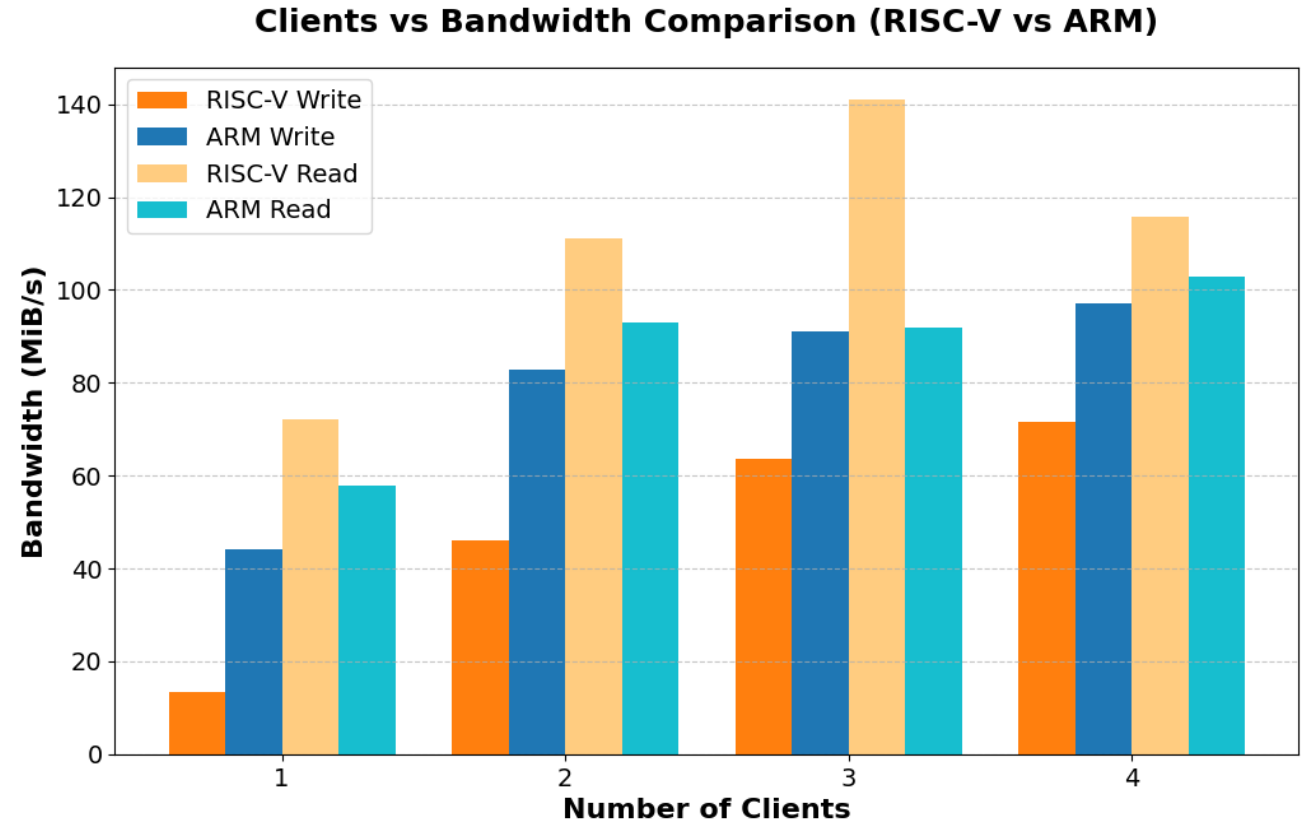
IOR – Multi Process Scaling

- **Configuration:** MPI processes scaled from 2 to 16 with a fixed 600 MiB block size and 128 KiB stripe size.
- **Observation:** Write bandwidth increases with process count on both RISC-V and ARM, while read bandwidth improves up to a peak and then stabilizes.
- **Conclusion:** Both systems exhibit good process-level scalability, with performance gains limited by system-level constraints at higher concurrency.



IOR – Client Count Scaling

- **Configuration:** beegfs client nodes scaled from **1 to 4**
- **Conclusion:** beegfs operates stably under increasing client parallelism on the RISC-V cluster



MD Test Results : RISC-V & ARM

```
-- started at 03/11/2026 12:06:05 --  
  
mdtest-4.0.0 was launched with 16 total task(s) on 4 node(s)  
Command line used: mdtest '-d' '/mnt/beegfs-share/ior/build/bin/mdtest.0' '-F' '-n' '1000' '-i' '3' '-u'  
Nodemap: 1111000000000000  
  
Path          : /mnt/beegfs-share/ior/build/bin/mdtest.0  
FS            : 456.3 GiB   Used FS: 0.0%   Inodes: 0.0 Mi   Used Inodes: nan%  
16 tasks, 16000 files  
  
SUMMARY rate: (of 3 iterations)  
  Operation      Max      Min      Mean      Std Dev  
  -----  
  File creation   1895.533  1850.233  1877.313    23.914  
  
  File stat       8180.344  7001.008  7614.980   591.168  
  
  File read       2990.518  2823.680  2896.595    85.380  
  
  File removal    1904.896  1813.545  1862.987    46.139  
  
  Tree creation    32.864    22.005    27.322     5.433  
  
  Tree removal     7.204     5.783     6.687     0.786  
  
-- finished at 03/11/2026 12:07:15 --
```

RISC-V

```
-- started at 03/25/2026 02:07:43 --  
  
mdtest-4.1.0+dev was launched with 16 total task(s) on 4 node(s)  
Command line used: mdtest '-d' '/mnt/beegfs-share/ior/build/bin/mdtest.0' '-F' '-n' '1000' '-i' '3' '-u'  
Nodemap: 1111000000000000  
  
Path          : /mnt/beegfs-share/ior/build/bin/mdtest.0  
FS            : 456.3 GiB   Used FS: 0.0%   Inodes: 0.0 Mi   Used Inodes: nan%  
16 tasks, 16000 files  
  
SUMMARY rate (in ops/sec): (of 3 iterations)  
  Operation      Max      Min      Mean      Std Dev  
  -----  
  File creation   2911.654  2559.785  2784.377    195.078  
  
  File stat       18370.936 17236.456 17909.050   595.869  
  
  File read       5694.471  5628.452  5658.453    33.418  
  
  File removal    3271.909  3135.663  3206.177    68.249  
  
  Tree creation    79.949    33.858    60.877     24.051  
  
  Tree removal    11.942    10.569    11.099     0.738  
  
-- finished at 03/25/2026 02:08:28 --
```

ARM

Key Observations

- Stable BeeGFS on RISC-V
 - BeeGFS demonstrates reliable metadata operation performance on native RISC-V hardware.
- Competitive Performance Trends
 - Similar metadata performance trends across RISC-V and ARM indicate the viability of RISC-V for HPC storage workloads.

IO500 Benchmark

- **Configuration:** IO500 executed on a 4-node cluster using 8 MPI processes.
- **Observation:** RISC-V achieved 0.065 GiB/s write and 0.059 GiB/s read in easy IOR tests, while metadata performance reached 1.33 kIOPS (easy) and 0.51 kIOPS (hard).
- **Conclusion:** Although the ARM cluster achieved a higher IO500 score (0.564 vs. 0.290), the RISC-V cluster demonstrated comparable performance trends.

```
[fedora@riscv03 io500]$ mpirun -np 8 --hostfile hostfile.txt ./io500 config-riscv.ini
IO500 version io500-sc25-5-g1d83996d7f05 (standard)
[RESULT] ior-easy-write 0.065456 GiB/s : time 327.716 seconds
[RESULT] mdtest-easy-write 1.331906 kIOPS : time 336.147 seconds
[ ] timestamp 0.000000 kIOPS : time 0.006 seconds
[RESULT] ior-hard-write 0.018425 GiB/s : time 474.448 seconds
[RESULT] mdtest-hard-write 0.510550 kIOPS : time 328.613 seconds
[RESULT] find 27.780200 kIOPS : time 22.097 seconds
[RESULT] ior-easy-read 0.059624 GiB/s : time 240.168 seconds
[RESULT] mdtest-easy-stat 6.139350 kIOPS : time 73.775 seconds
[RESULT] ior-hard-read 0.028944 GiB/s : time 170.334 seconds
[RESULT] mdtest-hard-stat 4.961377 kIOPS : time 34.787 seconds
[RESULT] mdtest-easy-delete 2.068692 kIOPS : time 224.633 seconds
[RESULT] mdtest-hard-read 0.819215 kIOPS : time 205.192 seconds
[RESULT] mdtest-hard-delete 0.602020 kIOPS : time 282.639 seconds
[ ] ior-rnd4K-easy-read 0.008010 GiB/s : time 362.496 seconds
[SCORE ] Bandwidth 0.037983 GiB/s : IOPS 2.218628 kiops : TOTAL 0.290292

The result files are stored in the directory: ./results/2026.03.09-10.39.22
```

```
[fedora@rpi3 io500]$ mpirun -np 8 --hostfile hostfile ./io500 config-riscv.ini
IO500 version io500-sc25-5-g1d83996d7f05 (standard)
[RESULT] ior-easy-write 0.107421 GiB/s : time 382.647 seconds
[RESULT] mdtest-easy-write 2.541319 kIOPS : time 310.588 seconds
[ ] timestamp 0.000000 kIOPS : time 0.003 seconds
[RESULT] ior-hard-write 0.048307 GiB/s : time 563.995 seconds
[RESULT] mdtest-hard-write 1.003141 kIOPS : time 301.725 seconds
[RESULT] find 61.109581 kIOPS : time 17.815 seconds
[RESULT] ior-easy-read 0.067792 GiB/s : time 609.496 seconds
[RESULT] mdtest-easy-stat 17.639521 kIOPS : time 45.639 seconds
[RESULT] ior-hard-read 0.057473 GiB/s : time 473.184 seconds
[RESULT] mdtest-hard-stat 15.992881 kIOPS : time 19.904 seconds
[RESULT] mdtest-easy-delete 2.986661 kIOPS : time 268.870 seconds
[RESULT] mdtest-hard-read 1.564585 kIOPS : time 193.821 seconds
[RESULT] mdtest-hard-delete 1.274626 kIOPS : time 240.380 seconds
[ ] ior-rnd4K-easy-read 0.028989 GiB/s : time 346.486 seconds
[SCORE ] Bandwidth 0.067055 GiB/s : IOPS 4.755968 kiops : TOTAL 0.564724

The result files are stored in the directory: ./results/2026.03.04-15.12.07
```


Elbencho Benchmark

- **Configuration:** Elbencho executed with 4 threads per node, 1 MB block size, 4 GB data per process.
- **Observation:** RISC-V achieved 80 MiB/s write and 99 MiB/s read throughput, while ARM achieved 111 MiB/s write and 56 MiB/s read throughput.
- **Conclusion:** RISC-V demonstrated competitive sequential I/O performance, outperforming ARM in read throughput while maintaining reasonable write performance.

```
[fedora@riscv03 bin]$ ./elbencho -w -r -t 4 -b 1M -s 4g -d --direct --sync --hostsfile hostfile.txt --csv aggregate_results.csv /mnt/beegfs-share/testing
```

OPERATION	RESULT TYPE	FIRST DONE	LAST DONE
-----	-----	-----	-----
SYNC	Elapsed time :	0ms	0ms

MKDIRS	Elapsed time :	8ms	12ms
	Dirs/s :	456	639
	Dirs total :	4	8

SYNC	Elapsed time :	0ms	0ms

WRITE	Elapsed time :	6m11.297s	6m44.876s
	Files/s :	0	0
	IOPS :	87	80
	Throughput MiB/s :	87	80
	Total MiB :	32594	32768
	Files total :	2	8

SYNC	Elapsed time :	0ms	0ms

READ	Elapsed time :	4m56.363s	5m30.417s
	Files/s :	0	0
	IOPS :	108	99
	Throughput MiB/s :	108	99
	Total MiB :	32167	32768
	Files total :	1	8

SYNC	Elapsed time :	0ms	0ms

```
[fedora@rpi3 bin]$ ./elbencho -w -r -t 4 -b 1M -s 4g -d --direct --sync --hostsfile hostfile --csv aggregate_results6.csv /mnt/beegfs-share/testing
```

OPERATION	RESULT TYPE	FIRST DONE	LAST DONE
-----	-----	-----	-----
SYNC	Elapsed time :	0ms	0ms

MKDIRS	Elapsed time :	2ms	4ms
	Dirs/s :	1540	1808
	Dirs total :	4	8

SYNC	Elapsed time :	0ms	0ms

WRITE	Elapsed time :	4m36.182s	4m52.775s
	Files/s :	0	0
	IOPS :	112	111
	Throughput MiB/s :	112	111
	Total MiB :	30952	32768
	Files total :	1	8

SYNC	Elapsed time :	0ms	0ms

READ	Elapsed time :	6m56.192s	9m42.604s
	Files/s :	0	0
	IOPS :	43	56
	Throughput MiB/s :	43	56
	Total MiB :	18147	32768
	Files total :	1	8

SYNC	Elapsed time :	0ms	0ms

Conclusion

- **Fully functional** BeeGFS parallel file system deployed on RISC-V HPC cluster (VisionFive2 boards)
- **IOR, MDTest, IO500 and elbencho** benchmarks show **reliable I/O** and **metadata** performance on low-power hardware
- Baseline comparison with ARM Raspberry Pi cluster validates **RISC-V competitiveness**
- RISC-V Suitable **for experimental workloads** and **future HPC system exploration**

Future Work

- Optimize the RISC-V software stack through **compiler tuning** and **kernel-level I/O** optimizations.
- Investigate high speed **interconnects**, **PCIe bandwidth** constraints and **storage interface** improvements to address system bottlenecks.
- Evaluate BeeGFS on **next-generation** RISC-V hardware and expand the cluster to a **larger number of nodes**.

Thank you!

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